

An Interactive VR-Based Simulation System for Household Electricity Consumption Estimation

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Abstract—Growing population and rising household appliance ownership in Indonesia have increased electricity demand, while public understanding of energy consumption patterns remains limited. Existing energy feedback is mostly static (e.g., monthly bills), and prior VR applications for residential design focus on visualization without energy simulation. This paper presents *ModXplore*, an interactive VR-based simulation system developed on Meta Quest 3 using Unity and the Agile Scrum method. The system enables users to explore and customize a virtual house, simulate household electricity consumption using a Bottom-Up Appliance Modeling approach, and automatically generate personalized appliance usage schedules via a Large Language Model (LLaMA 3.1 through the Groq API). Evaluation was conducted through black-box functional testing and a user study with 24 respondents using the System Usability Scale (SUS), User Experience Questionnaire (UEQ), and a Likert-scale questionnaire. All 29 functional test scenarios passed, the energy estimation module distinguished two usage scenarios with an 8.48% monthly cost difference, the average SUS score was 71.56 (Grade C+), and the UEQ average was 2.15 with five of six scales rated Excellent. Users also reported strong support for energy understanding and decision-making (average above 4.5). *ModXplore* demonstrates that immersive VR combined with appliance-level energy estimation and AI-assisted scheduling is capable of supporting household energy literacy and energy-related decision-making.

Index Terms—Virtual Reality, household electricity simulation, *ModXplore*, Bottom-Up Appliance Modeling, Large Language Model

I. INTRODUCTION

The continuous growth of Indonesia's population has driven a steady increase in the demand for housing. According to Statistics Indonesia (BPS), the national population growth rate has consistently ranged between 1% and 1.5% per year [1]. This growing demand requires prospective buyers and homeowners to consider multiple aspects of house design before making a decision, such as room layout, building materials, and electronic appliance configuration. To support this process, Virtual Reality (VR) has been used as an interactive medium for simulating and observing house designs, allowing users to obtain a more realistic impression of a house before it is built or purchased [2].

One important aspect to consider in house design simulation is the estimation of electricity consumption, since household appliance usage is a major contributor to a household's electricity demand [3], [4]. The growth of households has been accompanied by an increase in the number and variety of electronic appliances used, such as air conditioners, televisions, and refrigerators, as confirmed by the National Socioeconomic Survey (SUSENAS) [5]. This condition has caused household electricity demand to keep increasing alongside society's growing reliance on electronic appliances [6]. Nevertheless, public understanding of efficient energy consumption patterns remains limited [7].

A key barrier to energy-efficient behavior is the lack of accessible, actionable feedback on household energy use. Most users only receive aggregated information through monthly electricity bills, without understanding which appliances contribute most to their total consumption or how house design affects energy usage. Studies show that static, non-personalized energy feedback is insufficient to stimulate meaningful behavioral change [8]. Interactive and visual approaches have been shown to be more effective in raising energy literacy among building occupants [9], and gamification or simulation-based methods have demonstrated potential in motivating energy behavior change [10].

Virtual Reality (VR) has emerged as a powerful medium for interactive visualization, particularly in the architecture, engineering, and construction sectors [11]. In particular, Suryanto [2] developed a VR application for house design exploration and customization using Unity and Meta Quest 3. However, that system focused primarily on spatial visualization and aesthetic customization, without addressing electricity consumption simulation or energy estimation features. The integration of energy awareness into VR-based house design tools remains largely unexplored.

This paper presents *ModXplore*, an interactive VR-based simulation system developed for Meta Quest 3 using the Unity Engine. *ModXplore* allows users to explore and customize a virtual house while simultaneously simulating and estimating household electricity consumption using the Bottom-Up Appliance Modeling approach. The system further integrates

a Large Language Model (LLM)—specifically LLaMA 3.1 via the Groq API—to automatically generate personalized appliance usage schedules. The system was evaluated through black-box functional testing and a user study with 24 respondents using the System Usability Scale (SUS), User Experience Questionnaire (UEQ), and a Likert-scale questionnaire on energy understanding and decision-making support.

The main contributions of this work are: (1) an interactive VR simulation system integrating house design exploration with real-time, appliance-level electricity consumption estimation, supported by an AI-based schedule personalization feature that minimizes the manual configuration burden required from users; (2) an energy estimation module based on Bottom-Up Appliance Modeling that accurately distinguishes usage scenarios; and (3) an interactive decision-support medium for household energy efficiency, validated through functional testing and a comprehensive user study (SUS, UEQ, and a Likert-scale questionnaire) demonstrating strong usability and user-perceived support for energy-related decision-making.

II. RELATED WORK

A. VR for Building Visualization and Design

VR has been widely adopted in architecture and residential design to overcome the limitations of conventional 2D visualization. Lang and Zhuang demonstrated that combining VR with clustering algorithms for interior design improved modeling accuracy by over 20% compared to traditional SVM-based methods and enabled real-time evaluation of design changes [12]. Safikhani et al. confirmed that immersive VR environments significantly enhance users' spatial understanding by allowing first-person exploration of buildings, which is difficult to achieve with static renders [11]. Wang et al. further showed that scene-context-aware object interaction in VR reduces selection time and raises user satisfaction during interior configuration tasks [13]. The integration of VR with Building Information Modeling (BIM) has also been shown to improve simulation realism and design communication between users and architects [14], [15]. In the Indonesian context, Suryanto [2] developed a VR application for house design exploration and material customization on Meta Quest 3 using Unity; however, the system did not include any electricity consumption simulation or energy estimation feature.

B. VR and AR for Energy Simulation and Awareness

Beyond architectural visualization, VR and AR are increasingly applied to improve energy literacy and support energy-efficient behavior. Latini et al. developed an immersive VR training environment that integrates a residential Building Energy Model with real-time eco-feedback indicators; over 90% of participants showed improved knowledge of energy-related behavior after the session [16]. Maqbool et al. evaluated audio, visual, and social cues in VR for promoting energy conservation, finding that visual interventions produced statistically significant improvements in pro-environmental behavior scores [17]. Anifowose et al. introduced ENERGYSIM, a multi-user immersive VR prototype for building energy education

that allows users to explore the impact of building materials on energy consumption through interactive menu-driven walkthroughs [18]. In the AR domain, An et al. showed that overlaying real-time energy consumption data onto physical appliances via AR reduced energy use without degrading occupant comfort [19]. Frigura-Iliasa et al. proposed an interactive 3D simulator (ElectroHouse) connecting consumption graphs with appliance-level controls to improve household energy literacy [20].

Despite this progress, existing studies either focus on building-level energy modeling without appliance-level interactivity, or on energy awareness without integrating house design exploration. None of the reviewed systems combine immersive VR house customization, bottom-up appliance-level energy estimation, and AI-driven schedule personalization within a single platform, which is the gap this work addresses.

III. SYSTEM DESIGN

The overall application workflow begins when the user launches ModXplore and is presented with the main scene, where a house design is selected from the available models. The user is then transferred to the exploration scene, where two main paths are available: navigating between rooms within the selected house, or modifying the house appearance (materials, furniture, doors, windows, and lighting). After exploring and customizing the house, the user proceeds to the energy simulation stage, where appliance configurations are set—either manually or via the AI-based schedule personalization feature—and the resulting electricity consumption and cost estimates are displayed. This workflow ensures that house design exploration and energy simulation are presented as a single, continuous experience within the VR environment. Fig. 1 illustrates this overall application flow.

A. Development Methodology

ModXplore was developed using the Agile Scrum method, chosen for its iterative and adaptive nature, which suits VR application development where interaction flow, navigation comfort, interface readability, and energy simulation integration must be refined progressively. System requirements were organized into a product backlog covering virtual house visualization, user navigation, appliance interaction, usage-duration configuration, and energy estimation computation. Development proceeded through several one- to two-week sprints, each focused on a specific feature or component. Each sprint produced a product increment, which was tested internally before being reviewed in a sprint review session with the research supervisor. Feedback from each review was used to update the product backlog and prioritize subsequent sprints, ensuring the system met functional requirements prior to user evaluation.

B. System Architecture

ModXplore consists of three interconnected components. The *Unity VR Application* runs on Meta Quest 3 and serves as the primary user interface, rendering the interactive 3D

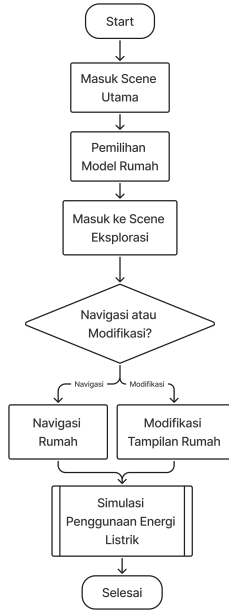


Fig. 1. Overall application workflow of ModXplore.

house environment and displaying real-time energy estimation. The *Node.js/Express.js* backend server acts as middleware, forwarding user activity descriptions to the LLM service via the Groq API. The *Python/FastAPI* speech-to-text server locally transcribes voice input using the Faster-Whisper model, enabling hands-free input within the VR environment. All three components communicate over a local Wi-Fi network using HTTP POST requests. The overall system architecture is illustrated in Fig. 2.

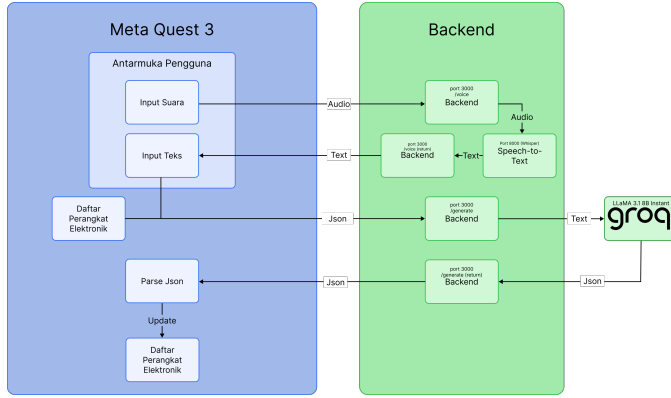


Fig. 2. ModXplore system architecture.

C. Energy Simulation Module

Energy estimation follows the Bottom-Up Appliance Modeling approach. Each virtual appliance is assigned a rated power P (W), quantity n , and user-configured usage duration t (hours). The energy consumed per appliance type per day is:

$$E = \frac{P \times t \times n}{1000} \quad (\text{kWh}) \quad (1)$$

Daily electricity cost is then estimated as:

$$B_{\text{daily}} = E_{\text{total}} \times T \quad (2)$$

where T is the applicable electricity tariff (Rp/kWh). Monthly cost accounts for 22 weekdays and 8 weekend days per month:

$$B_{\text{monthly}} = (B_{\text{weekday}} \times 22) + (B_{\text{weekend}} \times 8) \quad (3)$$

All parameters — power, quantity, start time, and duration — are configurable by the user through the in-VR interface. The simulation is estimative and does not account for real-time load variations or actual appliance efficiency losses. Fig. 3 shows the energy estimation panel as displayed inside the VR environment.



Fig. 3. Energy estimation display in the ModXplore VR environment.

D. AI-Based Schedule Personalization

To minimize the manual configuration burden required from users, ModXplore integrates LLaMA 3.1 8B Instant via the Groq API. Users describe their daily routine in natural language — either by text input or by voice transcribed locally via Faster-Whisper. A zero-shot prompt is constructed containing the activity description, the list of available virtual appliances, scheduling rules for weekdays and weekends, and a required JSON output schema. The LLM returns a structured schedule specifying each appliance's start time and usage duration for both day types. The Unity client parses the JSON and applies it to the simulation, instantly updating the energy and cost estimation display. Fig. 4 shows the AI-generated schedule result as presented to the user inside the VR application.

IV. EVALUATION

A. Functional Testing

Black-box testing was conducted across 29 test scenarios covering all major features: VR house exploration, room navigation, appliance interaction, schedule configuration, energy estimation display, voice input, and AI-generated schedule generation. Each scenario was verified against its expected



Fig. 4. AI-generated appliance schedule displayed in ModXplore.

output. Table I summarizes the test scenarios by functional category. All 29 scenarios passed without errors, confirming that the implemented features operate as designed within a single integrated VR environment.

TABLE I
FUNCTIONAL TESTING SUMMARY BY CATEGORY

Category	Scenarios	Passed
VR house exploration and customization (navigation, materials, furniture, doors/windows, lighting, appliance toggling)	18	18
Energy simulation and AI personalization (appliance configuration, energy/cost estimation, voice input, AI schedule generation)	11	11
Total	29	29

B. User Study

A user study was conducted with 24 respondents who tested the application as general users. Most respondents were aged 20–25 years (87.5%), with smaller groups aged 26–30 years (8.3%) and under 20 years (4.2%); 66.7% were male and 33.3% female, and 79.2% had prior experience using VR devices. Each participant completed a structured task scenario inside the VR environment, then filled out three evaluation instruments: the System Usability Scale (SUS), the User Experience Questionnaire (UEQ), and a Likert-scale questionnaire (1–5) assessing energy understanding and decision-making support.

The SUS score is computed from ten alternating positive and negative statements, each rated on a 0–4 scale, and converted to a 0–100 scale [21]:

$$SUS = 2.5 \times \sum_{i=1}^{10} x_i \quad (4)$$

where $x_i = \text{response}_i - 1$ for odd-numbered (positively worded) items and $x_i = 5 - \text{response}_i$ for even-numbered (negatively worded) items.

For UEQ, each of the six scales (Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty) is scored as the mean of its associated item ratings, where each item is

rated on a 7-point scale (1–7) and transformed to a –3 to +3 range [22]:

$$UEQ_{\text{scale}} = \frac{1}{n} \sum_{j=1}^n (r_j - 4) \quad (5)$$

where r_j is the raw rating of item j and n is the number of items belonging to that scale. The resulting scale means are then compared against the UEQ benchmark dataset.

V. RESULTS AND DISCUSSION

A. Functional Testing and Energy Estimation Results

All 29 functional test scenarios passed as designed, confirming that the implemented features operate correctly within a single integrated VR environment. For energy estimation accuracy, two appliance schedule configurations (Default and Energy-Saving) were compared. As shown in Table II, the Energy-Saving scenario yielded lower consumption on both weekdays and weekends, with a monthly cost difference of 8.48%, confirming that the Bottom-Up model correctly distinguishes usage patterns.

TABLE II
ENERGY ESTIMATION COMPARISON: DEFAULT VS. ENERGY-SAVING SCENARIO

Parameter	Default	Energy-Saving	Difference
Weekday consumption	13.40 kWh	12.38 kWh	1.02 kWh
Weekend consumption	14.44 kWh	12.90 kWh	1.54 kWh
Monthly cost	Rp592,728	Rp542,476	Rp50,252
Cost difference	—		8.48%

This difference originates from the appliance usage schedules underlying each scenario. Table III compares the Default schedule with the schedule generated by the LLM from the activity description "*I am a frugal person*" (Energy-Saving scenario). The AI-generated schedule reduced the usage duration of several appliances, particularly the air conditioner and bedroom lights, while leaving constantly-running appliances such as the refrigerator unchanged. The Bottom-Up Appliance Modeling computation (Eq. 1–3) directly reflects these reduced durations, which is why the Energy-Saving scenario in Table II consistently yields lower consumption and cost.

TABLE III
APPLIANCE SCHEDULE COMPARISON: DEFAULT VS. AI-GENERATED (ENERGY-SAVING)

Appliance	Default		AI-Generated	
	Weekday	Weekend	Weekday	Weekend
Refrigerator	00:00, 24h	00:00, 24h	00:00, 24h	00:00, 24h
Air Conditioner	22:00, 5h	21:00, 7h	22:00, 3h	21:00, 4h
Living Room Light	20:00, 4h	18:00, 6h	20:00, 4h	18:00, 5h
Bedroom 1 Light	20:00, 4h	18:00, 6h	21:00, 2h	20:00, 3h
Bedroom 2 Light	20:00, 4h	18:00, 6h	22:00, 2h	20:00, 3h
Bathroom Light	07:00, 1h	07:00, 1h	07:00, 1h	07:00, 1h

B. Usability and User Experience Results

A six-item Likert-scale questionnaire (Q1–Q6) was used to measure users’ understanding of the energy simulation feature, their perception of the AI scheduling feature, and the system’s support for decision-making. Results are shown in Table IV. All six items scored above 4.4 (Strongly Agree), with the hourly energy graph (Q3 = 4.71) and the AI scheduling feature (Q5 = 4.75) receiving the highest ratings, indicating that the visual representation of hourly consumption patterns and the AI-based schedule recommendation were particularly well-received by users. Fig. 5 shows the hourly energy consumption graph as displayed in the VR environment.

TABLE IV
USER EVALUATION QUESTIONNAIRE RESULTS (Q1–Q6)

Item	Statement	Mean
Q1	Helps understand appliance energy consumption	4.54
Q2	Gives a clear picture of electricity bill estimation	4.58
Q3	Hourly graph helps understand usage patterns	4.71
Q4	Helps consider energy efficiency when choosing appliances	4.50
Q5	AI feature eases configuration of an energy-saving schedule	4.75
Q6	Supports better decisions on electricity usage	4.54

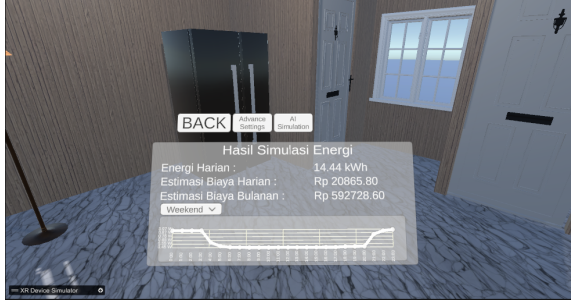


Fig. 5. Hourly electricity consumption graph in ModXplore.

The SUS evaluation yielded an average score of 71.56 (Grade C+, 60th–64th percentile on the Sauro–Lewis Curved Grading Scale), indicating an acceptable usability level. Individual scores ranged from 57.5 to 87.5, showing some variability in how easily respondents adapted to the VR interface. A Cronbach’s Alpha reliability test yielded a value of 0.54, below the commonly accepted threshold of 0.70 [23]. Analysis revealed that items Q4 (“need help from others”) and Q10 (“need to get used to the system”) had very low item-total correlations (0.036 and -0.005 , respectively). Distribution analysis showed that 70.8% of respondents agreed they needed to get accustomed to the system (Q10) while simultaneously rating the application as easy to use (Q3), suggesting that respondents interpreted “getting used to” as referring to the VR platform in general rather than the ModXplore application itself — consistent with the fact that 20.8% of respondents had no prior VR experience. This is noted as a limitation of the study.

The UEQ results per scale are shown in Table V. Five of six scales reached the *Excellent* benchmark; only *Perspicuity* was rated *Good* (1.91), reflecting that VR interaction was relatively unfamiliar for some respondents. This is consistent with the SUS reliability findings noted above, suggesting that initial VR learnability, rather than the energy simulation or AI scheduling features themselves, was the primary factor behind lower individual scores and reduced internal consistency.

TABLE V
UEQ EVALUATION RESULTS PER SCALE

Scale	Mean	Benchmark
Attractiveness	2.25	<i>Excellent</i>
Perspicuity	1.91	<i>Good</i>
Efficiency	2.16	<i>Excellent</i>
Dependability	2.05	<i>Excellent</i>
Stimulation	2.25	<i>Excellent</i>
Novelty	2.30	<i>Excellent</i>
Overall	2.15	—

All six items in Table IV achieved a mean score above 4.4, falling into the Strongly Agree category. These results indicate that users perceived the system as helpful in supporting energy understanding, AI-based schedule configuration, and energy-efficient decision-making.

C. Functional Comparison with Prior Work

ModXplore was developed as a continuation of the work by Suryanto [2], which also used Unity and Meta Quest 3 for VR house design exploration. Table VI compares the features available in both systems. ModXplore extends the prior foundation with three new capabilities — appliance-level energy simulation, daily/monthly kWh estimation, and AI-based schedule personalization — that were not available in the previous system.

TABLE VI
FEATURE COMPARISON WITH PRIOR WORK

Feature	Suryanto [2]	ModXplore
VR house exploration	✓	✓
Material customization	✓	✓
Furniture placement	✓	✓
Energy consumption simulation	×	✓
Daily/monthly kWh estimation	×	✓
AI schedule personalization	×	✓

The UEQ benchmark results for ModXplore are further illustrated in Fig. 6, showing that five of six scales achieved the *Excellent* category, with *Perspicuity* rated *Good* (1.91).

A few threats to validity should also be noted. First, the energy estimation comparison (Table II) was evaluated on a single house design (Type 36 Model 2) with a fixed appliance configuration; results may vary for other house types or appliance sets. Second, the Bottom-Up Appliance Modeling approach assumes constant rated power and does not account for real-time load variation, appliance efficiency degradation,

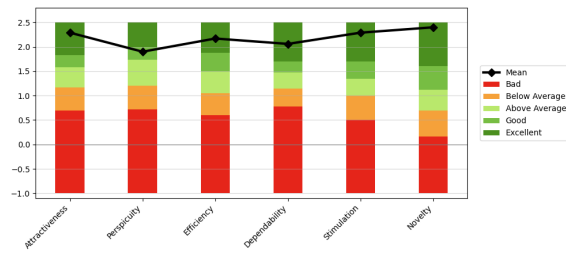


Fig. 6. UEQ benchmark results for ModXplore.

or standby consumption. Finally, the AI-based schedule personalization relies on an external LLM API (Groq), whose output quality and response time may vary depending on network conditions and prompt phrasing, which were not systematically varied in this study.

VI. CONCLUSION

This paper presented ModXplore, an interactive VR-based simulation system that integrates immersive house exploration, Bottom-Up Appliance Modeling for household electricity estimation, and LLM-based schedule personalization on Meta Quest 3. The system was evaluated through functional testing and a user study with 24 respondents. Results showed that all features performed correctly, the energy estimation module accurately differentiated usage scenarios (8.48% monthly cost difference), usability was acceptable (SUS = 71.56, Grade C+), and user experience was rated Excellent on five of six UEQ scales. Users also reported that the system supports energy understanding and decision-making (all Likert items above 4.4). A Cronbach's Alpha of 0.54 for the SUS instrument was observed, attributed to respondents' unfamiliarity with the VR platform rather than the application itself. Future work should address the limitations discussed above by evaluating the system with a larger and more diverse respondent group, validating the energy estimation against actual electricity consumption data, and testing across multiple house designs to improve generalizability.

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